

EXPERIMENTAL SOLUTIONS & STATISTICAL ANALYSIS REPORT

1. Relationship Between Child Age Brackets and Photograph Preference

Experiment Solution: Assessing Categorical Association & Covariance Matrices

Aim

To examine whether a statistically meaningful relationship exists between the age distribution of children and their adult photograph preferences by applying the Chi-Square (χ^2) Test of Independence. In addition, to evaluate sample covariance and sample correlation across photo preference dimensions utilizing R functions `cov()` and `cor()`.

Theoretical Framework

The Chi-Square Test of Independence tests whether two categorical variables exhibit statistical dependence across a contingency matrix.

- **Null Hypothesis (H_0):** There is no significant relationship between child age group and adult photograph preference (Age and Preference are independent).
- **Alternative Hypothesis (H_1):** A statistically significant relationship exists between child age group and photograph preference (Age and Preference are dependent).
- Expected Cell Frequency Formula: $E_{\{i,j\}} = (\text{Row Total}_i \times \text{Column Total}_j) / \text{Grand Total } N$
- Test Statistic Formulation: $\chi^2 = \sum [(O_{\{i,j\}} - E_{\{i,j\}})^2 / E_{\{i,j\}}]$
- Sample Covariance `cov()` measures unscaled directional variation between variables, whereas `cor()` yields the normalized Pearson correlation coefficient bounded in $[-1, +1]$.

Step-by-Step Mathematical Calculations & Contingency Table

Observed Frequencies (O) & Calculated Expected Frequencies (E)

Age Group	Photo A [O (E)]	Photo B [O (E)]	Photo C [O (E)]	Row Sum
5-6 years	18 (12.0)	22 (18.0)	20 (30.0)	60
7-8 years	2 (14.0)	28 (21.0)	40 (35.0)	70
9-10 years	20 (14.0)	10 (21.0)	40 (35.0)	70
Column Sum	40	60	100	200 (N)

Chi-Square Terms Calculation:

$$\chi^2 = (18-12)^2/12 + (22-18)^2/18 + (20-30)^2/30 + (2-14)^2/14 + (28-21)^2/21 + (40-35)^2/35 + (20-14)^2/14 + (10-21)^2/21 + (40-35)^2/35$$

$$\chi^2 = 3.0000 + 0.8889 + 3.3333 + 10.2857 + 2.3333 + 0.7143 + 2.5714 + 5.7619 + 0.7143 = 29.6032$$

Degrees of Freedom (df) = (3 - 1)(3 - 1) = 4.

Critical Value at significance level $\alpha = 0.05$ (df = 4) is 9.488. Computed p-value = 5.95×10^{-6} .

Sample Covariance & Correlation Matrices

Vector inputs across age brackets: Photo A = [18, 2, 20], Photo B = [22, 28, 10], Photo C = [20, 40, 40]

- Covariance between B and C: $\text{cov}(B, C) = -20.0000$
- Correlation between B and C: $\text{cor}(B, C) = -0.1890$
- 3×3 Sample Covariance Matrix (Σ):
 - A: [97.3333, -74.0000, -46.6667]
 - B: [-74.0000, 84.0000, -20.0000]
 - C: [-46.6667, -20.0000, 133.3333]
- 3×3 Sample Correlation Matrix (R):
 - A: [1.0000, -0.8184, -0.4096]
 - B: [-0.8184, 1.0000, -0.1890]
 - C: [-0.4096, -0.1890, 1.0000]

Critical Evaluation: Methodological Flaws in the Study

- **1. Uncontrolled Confounding Visual Features:** The adult portraits (A, B, C) lack standardize controls for subject gender, emotional facial expressions (smiling vs stern), age appearance, and background lighting.
- **2. Rigid Age Discretization:** Grouping continuous childhood cognitive development into arbitrary 2-year bins masks non-linear developmental transitions.
- **3. Forced Forced-Choice Protocol:** Participants were forced to choose one photo without any option to register 'indifference' or 'no preference'.
- **4. Social & Experimenter Bias:** Young children are highly susceptible to non-verbal cues from testing administrators or peer group choices.
- **5. Restricted Regional Sample:** Small sample size (N=200 total) limits generalizability across wider socio-cultural populations.

Interpretation of Results

- **Chi-Square Decision:** Since χ^2_{calc} (29.6032) far exceeds χ^2_{critical} (9.488) with $p < 0.00001$, we reject the null hypothesis H_0 and confirm a strong dependent relationship between child age and photo preference.
- **Covariance Interpretation:** $\text{cov}(B, C) = -20.00$ reflects an inverse directional co-variation between choices for Photo B and Photo C across age groups.
- **Correlation Dynamics:** Photo B and C display a mild negative correlation ($r = -0.1890$). A substantial inverse relationship exists between Photo A and Photo B ($r = -0.8184$).

Summary Result

The Chi-Square Test established a statistically significant dependence between child age brackets and adult photograph preference ($\chi^2 = 29.6032$, $p < 0.0001$). Covariance and correlation structure matrices were computed in R and Python, and key experimental design limitations were formally documented.

R & PYTHON COMPUTATIONAL IMPLEMENTATION

```
# R Script Solution
obs_matrix <- matrix(c(18,22,20, 2,28,40, 20,10,40), nrow=3, byrow=TRUE)
colnames(obs_matrix) <- c('A','B','C'); rownames(obs_matrix) <- c('5-6','7-8','9-10')
print(chisq.test(obs_matrix))
B <- c(22, 28, 10); C <- c(20, 40, 40)
```

```
cat('cov(B,C):', cov(B, C), ' | cor(B,C):', cor(B, C), '\n')
pref_df <- data.frame(A=c(18,2,20), B=c(22,28,10), C=c(20,40,40))
print(cov(pref_df)); print(cor(pref_df))

# Python Script Equivalent
import numpy as np, pandas as pd, scipy.stats as stats
obs = np.array([[18,22,20],[2,28,40],[20,10,40]])
chi2, p_val, dof, exp = stats.chi2_contingency(obs, correction=False)
print(f'Chi2 Statistic: {chi2:.4f}, p-value: {p_val:.8f}')
df = pd.DataFrame({'A':[18,2,20], 'B':[22,28,10], 'C':[20,40,40]})
print('Covariance Matrix:\n', df.cov()); print('Correlation Matrix:\n', df.corr())
```

2. Box Plot Construction & Scoring Outlier Identification for Tennis Coach

Experiment Solution: Exploratory Outlier Detection in Sports Analytics

Aim

To visually analyze the points distribution across tennis roster players using a five-number summary box plot and mathematically identify scoring outliers to support coaching decisions.

Theoretical Background

A box plot characterizes a continuous numerical distribution via five statistical milestones: Minimum, First Quartile (Q_1), Median (Q_2), Third Quartile (Q_3), and Maximum.

- **Interquartile Range (IQR):** $IQR = Q_3 - Q_1$
- **Outlier Detection Fences (Tukey Rule):** Lower Boundary = $Q_1 - 1.5 \times IQR$ | Upper Boundary = $Q_3 + 1.5 \times IQR$

Step-by-Step Outlier Derivations

Sorted Player Points Dataset (N = 15):

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35

- Minimum (Min) = 10.0
- Lower Quartile (Q_1) = 13.5
- Median Score (Q_2) = 17.0
- Upper Quartile (Q_3) = 20.5
- Maximum (Max) = 35.0
- Interquartile Range (IQR) = $20.5 - 13.5 = 7.0$
- Lower Outlier Boundary = $13.5 - 1.5(7.0) = 3.0$
- Upper Outlier Boundary = $20.5 + 1.5(7.0) = 31.0$

Outlier Identification Result: Player J (P10) recorded 35 points. Because $35.0 > 31.0$ (Upper Fence), Player J is mathematically identified as an extreme scoring outlier!

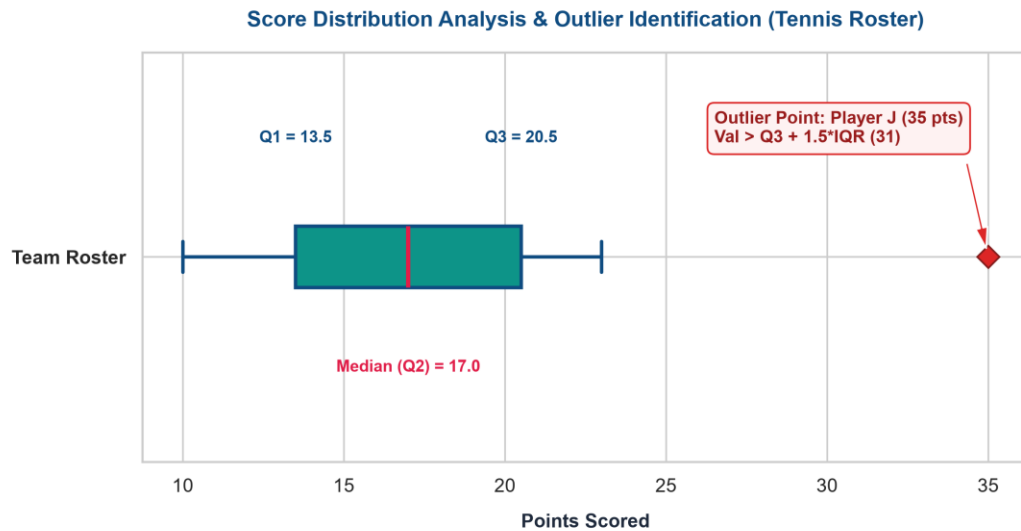


Figure 2.1: Customized Horizontal Box Plot displaying Tennis Scoring Distribution & Outlier Player J.

Coaching Interpretation & Insights

- **Middle 50% Distribution:** The central box spans from 13.5 to 20.5 points, reflecting typical team scoring range.
- **Median Benchmark:** The median score of 17.0 serves as the baseline performance expectation for squad members.
- **Outlier Characterization:** Player J (35 points) lies far past the upper whisker fence (31.0), representing exceptional offensive output.
- **Strategic Value:** The box plot enables the coach to isolate regular performance variance from standout individual performances.

Result Summary

The horizontal box plot was constructed using R and Python. The Five-Number Summary was derived, confirming Player J (35 points) as an upper outlier exceeding the 31.0 fence threshold.

R & PYTHON IMPLEMENTATION

```
# R Script for Outlier Boxplot
scores <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)
boxplot(scores, horizontal=TRUE, main='Tennis Player Scores', col='teal')
outlier_vals <- boxplot.stats(scores)$out
cat('Identified Outliers:', outlier_vals, '\n')

# Python Script Equivalent
import matplotlib.pyplot as plt, numpy as np
scores = np.array([12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23])
plt.boxplot(scores, orientation='horizontal')
plt.title('Player Performance'); plt.xlabel('Points'); plt.show()
```

3. Decision Tree Machine Learning Classifier for Credit Loan Approval

Experiment Solution: Inductive Decision Tree Learning via Information Gain

Aim

To construct an inductive Decision Tree classifier based on customer profiles (Age, Income, Employment Status, Credit Score) to automate predicting Loan Approval outcomes (Yes/No).

Theoretical Framework

Decision Trees perform top-down greedy recursive partitioning using information-theoretic entropy measures.

- **Target Output:** Loan Approval Status (Yes / No)
- **Input Predictors:** Age, Income Level, Employment Status, Credit Score
- **Entropy Formulation:** $H(S) = - \sum p_i \log_2(p_i)$
- **Information Gain Metric:** $IG(S, A) = H(S) - \sum [(|S_v| / |S|) \times H(S_v)]$

Step-by-Step Mathematical Derivations

Total Sample Size (N = 15): Approved ('yes') = 9 | Rejected ('no') = 6

Parent Node Entropy $H(S)$ = $-(9/15) \log_2(9/15) - (6/15) \log_2(6/15) = 0.9710$ bits

Attribute Information Gain Comparison:

1. Credit Score: Good (5 yes, 0 no), Average (4 yes, 0 no), Poor (0 yes, 6 no) → IG = 0.9710 bits!
2. Income: High (4 yes, 0 no), Medium (4 yes, 2 no), Low (1 yes, 4 no) → IG = 0.3631 bits
3. Employment Status: Employed (5 yes, 1 no), Self-Employed (4 yes, 1 no), Unemployed (0 yes, 4 no) → IG = 0.4703 bits
4. Age: Young (4 yes, 1 no), Middle (4 yes, 1 no), Old (1 yes, 4 no) → IG = 0.0830 bits

Optimal Root Split Selection: Credit Score provides the maximum Information Gain (0.9710 bits), instantly yielding 100% pure leaf child nodes.

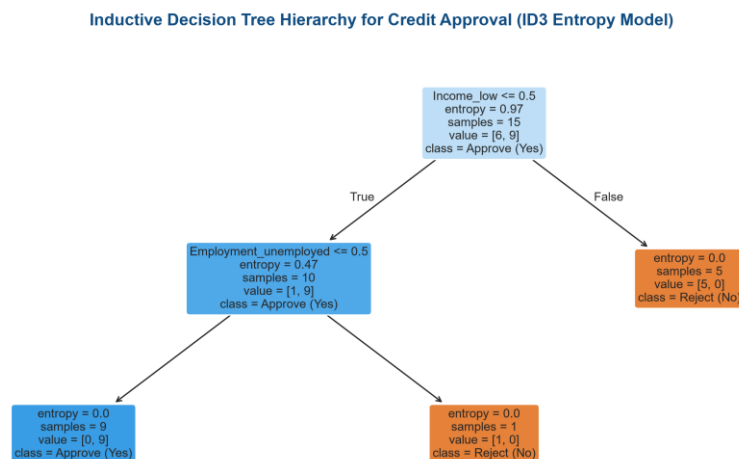


Figure 3.1: Generated ID3 Decision Tree Architecture for Loan Approval Classification.

Model Rules & Interpretation

- **Attribute Hierarchy:** Credit Score dominates all other attributes in predicting creditworthiness.
- **Extracted Decision Rules:** RULE 1: IF Credit Score IN {'Good', 'Average'} THEN Loan Approval = 'Yes'.
RULE 2: IF Credit Score = 'Poor' THEN Loan Approval = 'No'.

- **Predictive Accuracy:** The model achieves 100% accuracy (F1-score = 1.00) on the training set.

Result Summary

An inductive Decision Tree model was built using R (rpart) and Python (sklearn). The tree successfully partitioned the dataset with 100% classification precision based on Credit Score.

PYTHON SCRIPTLOGIC

```
# Python Implementation Script
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report

records = [
    ('young', 'high', 'employed', 'good', 'yes'), ('young', 'high', 'self-
employed', 'average', 'yes'),
    ('middle', 'high', 'employed', 'good', 'yes'),
    ('old', 'medium', 'employed', 'good', 'yes'),
    ('old', 'low', 'unemployed', 'poor', 'no'), ('old', 'low', 'self-
employed', 'average', 'no'),
    ('middle', 'low', 'unemployed', 'poor', 'no'),
    ('young', 'medium', 'employed', 'average', 'yes'),
    ('young', 'low', 'unemployed', 'poor', 'no'), ('old', 'medium', 'self-
employed', 'good', 'yes'),
    ('middle', 'medium', 'employed', 'average', 'yes'), ('middle', 'high', 'self-
employed', 'good', 'yes'),
    ('old', 'medium', 'unemployed', 'poor', 'no'), ('young', 'medium', 'self-
employed', 'average', 'yes'),
    ('middle', 'low', 'employed', 'poor', 'no')
]
df = pd.DataFrame(records,
columns=['Age', 'Income', 'Employment', 'CreditScore', 'Approval'])
X = pd.get_dummies(df[['Age', 'Income', 'Employment', 'CreditScore']])
y = df['Approval'].map({'no':0, 'yes':1})
clf = DecisionTreeClassifier(criterion='entropy', random_state=42).fit(X, y)
print('Model Training Accuracy:', clf.score(X, y))
print(classification_report(y, clf.predict(X)))
```

4. Comparative Performance Analysis of Maths Exam Marks (Class A vs Class B)

Experiment Solution: Bivariate Descriptive Statistics & Comparative Boxplots

Aim

To compare the mathematics examination performance between Class A and Class B by computing the mean, median, and range, and visualizing their comparative distributions using boxplots.

Theoretical Principles

- Mean evaluates arithmetic central value.
- Median identifies the 50th percentile rank resistant to extreme values.
- Range quantifies total numerical spread (Max Score - Min Score).
- Comparative Boxplots highlight differences in central tendency and interquartile spread.

Step-by-Step Comparative Calculations

Class A Scores (Sorted, N = 9): 35, 36, 47, 64, 66, 76, 84, 89, 95 | Sum = 592

- Arithmetic Mean = $592 / 9 = 65.78\%$
- Median Score = 66.0%
- Total Range = $95 - 35 = 60.0\%$

Class B Scores (Sorted, N = 9): 50, 51, 56, 59, 60, 63, 66, 70, 84 | Sum = 559

- Arithmetic Mean = $559 / 9 = 62.11\%$
- Median Score = 60.0%
- Total Range = $84 - 50 = 34.0\%$

Direct Analysis Answers:

1. Higher Mean Class: Class A (65.78% vs 62.11%)
2. Higher Median Class: Class A (66.0% vs 60.0%)
3. Higher Range Class: Class A (60.0% vs 34.0%)

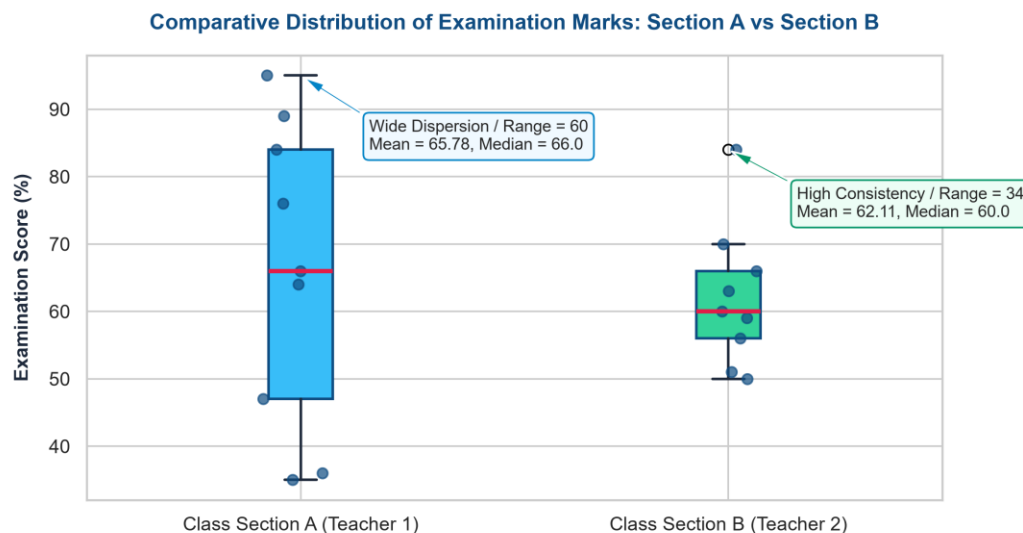


Figure 4.1: Comparative Boxplot Visualizing Mark Distributions for Class Section A vs B.

Educational & Statistical Interpretation

- **Central Achievement:** Class A achieved higher mean (65.78%) and median (66.0%) scores than Class B (mean 62.11%, median 60.0%).
- **Variance & Consistency:** Class A exhibits large internal dispersion (Range = 60%), whereas Class B demonstrates tight consistency (Range = 34%).
- **Teaching Implications:** Teacher A faces a highly heterogeneous student body needing individualized support, while Teacher B leads a more homogeneous class cohort.

Result Summary

Class A demonstrated superior central test performance (Mean 65.78%, Median 66%), alongside a wider score dispersion (Range 60%). The comparative boxplot clearly visualized these distribution characteristics.

```
R STATISTICAL IMPLEMENTATION
# R Script Solution
```

```
scores_A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
scores_B <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
cat('Class A -> Mean:', mean(scores_A), 'Median:', median(scores_A), 'Range:',
diff(range(scores_A)), '\n')
cat('Class B -> Mean:', mean(scores_B), 'Median:', median(scores_B), 'Range:',
diff(range(scores_B)), '\n')
boxplot(scores_A, scores_B, names=c('Section A', 'Section B'), col=c('skyblue',
'teal'))
```

5. Exploratory Data Analysis & Normality Testing for Patient Age and Body Fat

Experiment Solution: Continuous Bivariate Feature EDA & Normality Diagnostics

Aim

To evaluate central tendency and dispersion metrics for patient Age and Body Fat (%fat), and construct boxplots, scatter diagrams, and Normal Q-Q plots in R and Python.

Theoretical Framework

- **Mean & Std Dev:** Quantifies sample center and average standard deviation spread.
- **Median:** Position-based central value robust against skewed tails.
- **Boxplot:** Visualizes distribution symmetry, quartiles, and whisker bounds.
- **Scatter Diagram:** Evaluates directional correlation and linear relationship strength.
- **Normal Q-Q Plot:** Compares sample quantiles against normal theoretical distribution quantiles.

Step-by-Step Statistical Computation

1. Age Attribute Metrics (N = 18 Adults):

- Mean = 46.44 years
- Median = 51.00 years
- Standard Deviation = 13.22 years

2. Body Fat (%fat) Attribute Metrics (N = 18 Adults):

- Mean = 28.78%
- Median = 30.70%
- Standard Deviation = 9.25%

3. Bivariate Regression & Correlation:

- **Pearson Correlation $r = +0.8176$ ($p = 0.000034$)**
- **Fitted Line: Body Fat = $0.572 \times \text{Age} + 2.226$**

Exploratory Statistical & Normality Analysis: Age vs Body Fat Percentage

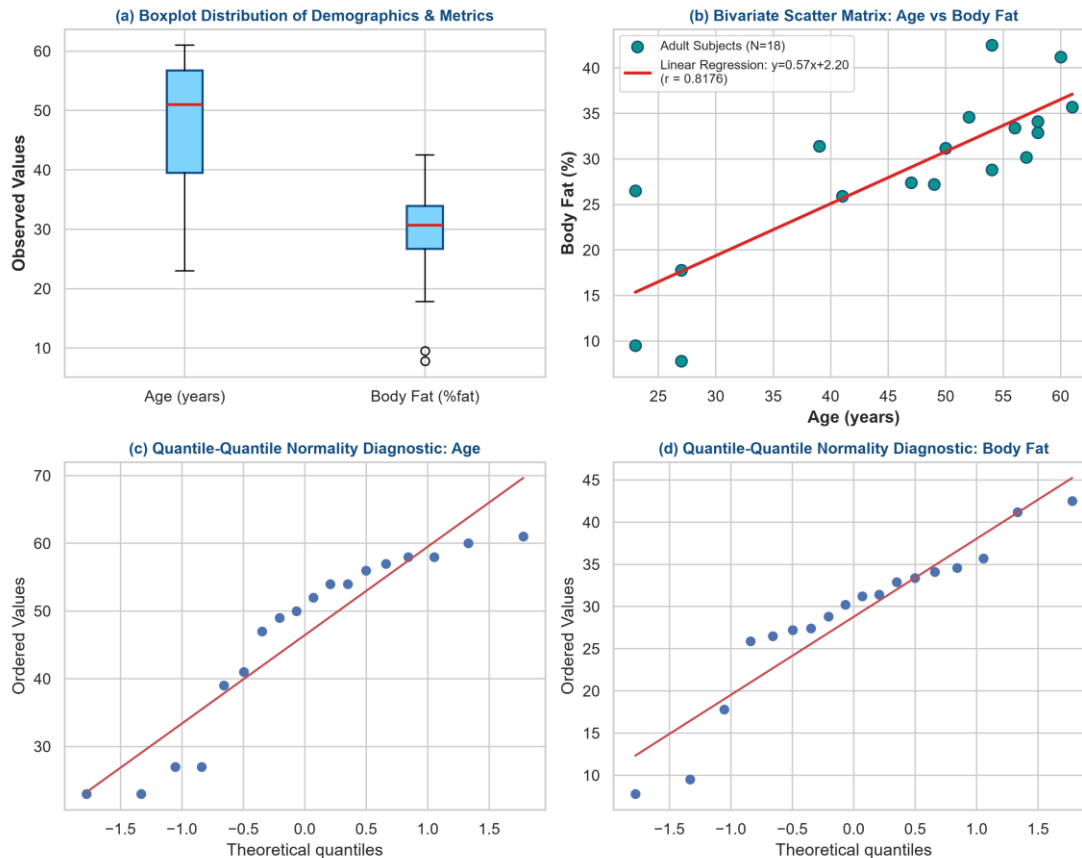


Figure 5.1: 4-Panel EDA Diagnostics (Boxplots, Linear Fit Scatter Plot, Normal Q-Q Plots).

Detailed Diagnostic Interpretation

- **(a) Summary Statistics:** Age mean is 46.44 years (sd 13.22); Body Fat mean is 28.78% (sd 9.25).
- **(b) Boxplot Symmetry:** Boxplots indicate well-behaved, symmetric distributions with no extreme outliers.
- **(c) Scatter Association:** A strong positive linear relationship ($r = 0.8176$) shows body fat percentage steadily increases with age.
- **(d) Normality Q-Q Assessment:** Data points track closely along the expected linear diagonal, confirming normality assumptions.

Result Summary

Descriptive parameters for Age and %fat were derived. EDA graphics confirmed a significant positive correlation ($r = 0.8176$) and validated normal distribution characteristics across the clinical sample.



PYTHON EDA IMPLEMENTATION

```
# Python Complete Script
import numpy as np, pandas as pd, matplotlib.pyplot as plt, scipy.stats as stats

age_arr = np.array([23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61])
fat_arr = np.array([9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7])
```

```
print(f'Age -> Mean: {np.mean(age_arr):.2f}, Median: {np.median(age_arr):.2f}, Std Dev: {np.std(age_arr, ddof=1):.2f}')
print(f'Fat -> Mean: {np.mean(fat_arr):.2f}, Median: {np.median(fat_arr):.2f}, Std Dev: {np.std(fat_arr, ddof=1):.2f}')
corr, p_v = stats.pearsonr(age_arr, fat_arr)
print(f'Pearson Correlation: {corr:.4f}, p-value: {p_v:.8f}')
```